UNIVERSITY OF TEHRAN
Electrical and Computer Engineering Department
Digital Logic Design, ECE 367 / Digital Systems I, ECE 894
Spring 1404
Computer Assignment 2
Basic Switch and Gate Structures in SystemVerilog
Week 6

Name:	
Date:	

1. Part 5 of Computer Assignment 1 is the GT output of a 2-bit comparator. Describe this circuit using an **assign** statement considering delays back annotated from the corresponding transistor-level circuit.

2. Design a gate-level circuit for describing EQ (equal) output of a 2-bit comparator. Describe this circuit using an **assign** statement considering delays back annotated from the corresponding transistor-level circuit.

3. Using Part 1 and Part 2, above, describe a 2-bit comparator with GT and EQ outputs. Generate a testbench and test this circuit for several input values. In the testbench use **repeat**, **$random**, add and other arithmetic operations for generation of data. Find the worst-case delay of this circuit.

4. Cascading the two 2-bit comparator circuits of Part 3 into a 4-bit comparator can be achieved by a circuit that takes a set of GT EQ outputs from the most significant two bits (GT_1, EQ_1), a set of GT EQ outputs from the least significant two bits (GT_0, EQ_0), and generates GT and EQ for the 4-bit comparator. Build this circuit using gates from Computer Assignment 1. Describe this circuit using an **assign** statement considering delays back annotated from the corresponding transistor-level circuit. We refer to this circuit as GEcascade.

5. Using two 2-bit comparators and a GEcascade, build a 4-bit comparator. Use instantiation statements. Generate a testbench and test this circuit for several input values. In the testbench use **repeat**, **$random**, add and other arithmetic operations for generation of data. Find the worst-case delay of this circuit.

6. In a tree structure build a 16-bit comparator with 2-bit comparators in the first layer and GEcascade circuits for the other layers. Use instantiation statements. Generate a testbench and test this circuit for several input values. In the testbench use **repeat**, **$random**, add and other arithmetic operations for generation of data. Find the worst-case delay of this circuit.

Hint: Built a 4-bit comparator, then instantiate two 4-bit comparators and a GEcascade to build an 8-bit comparator, then instantiate two 8-bit comparators and a GEcascade to build a 16-bit comparator.

7. Show the design of an n-bit comparator where n is 2^i (i>1). Use Verilog **generate** statements. In the testbench use **repeat**, **$random**, add and other arithmetic operations for generation of data. Find the worst-case delay of this circuit.

Deliverables:

Generate a report that includes items discussed below for each parts of this CA. Do not show any code without first showing its circuit diagram.

A. Show the circuit diagram that you are analyzing. Show gates and/or transistors according to the specified delays.
B. Hand-simulate or analyze the circuits you have shown in Parts 3, 5, 6, and 7 and write your expected values. For example, indicate what values you expect for the worst-case delays and express your reason for that.
C. Show your SystemVerilog descriptions of the design you are simulating and the testbench for it. Best is to use the Snip tools to get the image from Notepad++. This way you see all keywords and indentations. Make sure your SV codes are properly indented and all line-up rules are followed.
D. Show an image of the project that you have created in ModelSim for the simulation of your circuit.
E. When simulations are complete, or even if you have a partial simulation, include an image of the output waveforms showing signal names being displayed.

Make a PDF file of your report and name it with the format shown below:

FirstinitialLastnameStudentnumber-CAn-ECEmmm

Where *nn* is a two-digit number for the Computer Assignment, *mmm* is the three-digit course number under which you are registered, and hopefully you know the rest. For the *Firstinitial* use only one character. For *Lastname* and for the multi-part last names use the part you are most identified with. Use the last five digits of your student id (exclude 8101) for the *Studentnumber* field of the report file name.